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RESEARCH ARTICLE

Ontology-Based Neuro-Symbolic AI: Effects on Prediction Quality and Explainability

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ABSTRACT Artificial intelligence (AI) systems, based on neural networks are becoming ubiquitous. However, in many cases, their application is constrained by the lack of interpretability. One of the ways of overcoming this limitation is neuro-symbolic approaches, where the neural network is complemented by some symbolic structures, expressing existing knowledge and making it understandable to humans. Such approaches can make predictions more understandable and interpretable (due to the connection to human-understandable symbolic structures). Still, they can also have an additional advantage by allowing models to be trained with less data (due to leveraging prior knowledge). This paper focuses on a subset of neuro-symbolic approaches, where domain ontologies play the role of symbolic structures. The paper discusses the existing methods to build ontology-aware explainable neural networks and ways to leverage ontologies in forming the explanation. Based on the analysis, it proposes a computational framework for building ontology-aware self-explaining neural networks. The proposed framework allows several specializations, and it is shown that these specializations allow one to improve the prediction quality. Finally, the paper presents the results of a user study, showcasing that ontology-based explanations can improve the understandability of an AI model and the efficiency of human-AI interaction.

INDEX TERMS Deep learning, explainable AI, machine learning, neural networks, ontology, symbolic AI.

I. INTRODUCTION

Over the past decade, the utilization of neural networks in Artificial Intelligence (AI) has proliferated significantly. Artificial neural networks have emerged as pivotal tools employed to address an extensive array of information processing challenges, with particular emphasis on semi-structured data domains such as video, audio, and textual data. At the same time, one of the significant disadvantages of the neural network approach to AI in decision-making (compared to the symbolic AI) is that the result of the neural network is not always easily interpretable and explainable. The lack of explainability limits experts' confidence in the results obtained from neural networks and

hinders their use in critical areas. This problem is recognized by the scientific community – to date, a wide range of methods have been proposed aimed at interpreting and explaining predictions obtained using neural networks (see, for example, review [1]). However, the vast majority of these methods are mostly intended for experts in the field of machine learning and artificial intelligence, and not for problem domain experts [2].

One pivotal domain where the deficiency in explainability of conventional neural-based systems poses a significant challenge is within collaborative decision support systems, which rely on human-AI interaction. In such systems, a heterogeneous team (consisting of human experts and AI agents) collaborates to address a certain problem specified by the end user. This collaboration involves gathering and processing information, as well as generating and assessing potential

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solutions, thereby empowering the end user to make well-informed decisions [3], [4], [5].

The basis of collaborative processes is the exchange of knowledge and mutual learning (the transfer of knowledge from a person to an AI agent and vice versa). Such knowledge transfer typically requires the presence of some symbolic system that ensures interaction [6] (e.g., language, understandable by both parties).

The use of symbols as the basis of collaborative processes, on the one hand, and the spread of neural network AI, on the other, necessitate the synthesis of neural network and symbolic paradigms in relation to the creation of collaborative decision support systems. Such a synthesis is called neuro-symbolic artificial intelligence [7]. Neuro-symbolic artificial intelligence refers to a very wide range of methods focused on bringing together the neural and symbolic traditions in AI (see, e.g., [8], [9] for a detailed review). Several alternative spellings are currently used (neurosymbolic, neural-symbolic), but following [10] we consider them identical, opting for the “neuro-symbolic”. Such methods have become widespread in many application domains and are used both to improve the quality of prediction and the interpretability of AI models. For example, neuro-symbolic AI is active in fields such as medicine ([11], [12]), education ([13], [14]), robotics ([15], [16]), and many others. In this paper, we focus on one specific symbolic representation, namely, ontologies, besides, the discussed methods fall mostly into the group: “cascading from a neural system into a symbolic reasoner”, as introduced in the high-level classification used in [10]. The significance of employing ontologies as the foundation of a communication system for collaborative decision support stems from two key factors:

- 1) Ontologies serve as formalized representations of problem-specific terminology, providing a uniform and comprehensible language for experts and enabling automated processing (logical inference).
- 2) A vast array of ontologies has been developed across various problem domains, documented using standardized such as OWL [17]. This wealth of knowledge holds the potential for widespread utilization across diverse applications.

Specifically, the paper considers the following scenario. There is a binary classification problem, i.e., for each input sample the model has to decide if the sample belongs to the given target class. There is also a domain ontology, defining concepts that can be used to describe input samples (including the target class). This paper examines the potential of ontology-based neuro-symbolic AI in three ways:

- 1) The paper proposes a computational framework to embed domain ontology into the neural network during training. It can help, first, to improve the prediction quality of the network, and, second, to achieve one of the forms of explainability of the neural network – in particular, ontology-based explainability.

- 2) Based on the proposed framework, the paper explores several specific neural network training scenarios, adapted to various situations (some scenarios are suited to the situation when there are many samples labeled only on target class, while others can leverage auxiliary labels with ontology concepts other than target class or even lack of labels).
- 3) Finally, the paper examines various ways of presenting ontology-based explanations and explores their understandability.

However, it is essential to acknowledge that ontology-based approaches to explaining neural networks can be categorized with respect to other explanation methodologies in the following manner. Every attempt to achieve explainability hinges on the notion that at a certain level of abstraction, further elucidation becomes unnecessary [18]. In this context, ontology-based methods and “classical” approaches (such as proxy-based or attribution-based methods) occupy distinct niches, each selecting different levels of abstraction where explanations are deemed unnecessary. For example, proxy-based methods [19], [20], [21] (in which the behavior of the explained model is imitated by a simpler interpreted model) may distort the representation of the original model and present difficulties for human understanding. Attribution-based methods (e.g., [22], [23]) provide information about which input features influenced the classification result, however, these methods do not provide any information about why this is the case. Thus, the interpretation of the model result is still left to the person.

In many ontology-based techniques, the neural network task is conceptualized as comprising two primary subtasks: 1) the transformation from raw observations (data) to concepts (symbols), and 2) the derivation of the final outcome through manipulation of these concepts (symbols). Here, the focus of explanation lies on the symbolic aspect, while the first subtask — the transition to symbols — is typically considered self-explaining. Conversely, classical methods, more prevalent in practice, often overlook the presence of a symbolic component and base explanations solely on the initial features of the task (i.e., neural network inputs).

This distinction yields two notable consequences. Firstly, the assumption regarding the existence of two subtasks imposes constraints on the applicability of ontology-based methods; their effectiveness relies on the feasibility of representing the problem solution in symbolic terms, necessitating an adequate set of symbols to express the target variable. Secondly, ontology-based and classical methodologies can effectively complement each other, offering explanations at varying levels of abstraction.

There are several approaches for incorporating logical constraints into neural networks (NeurASP [24], DeepProbLog [25], LTN [26], STE [27], and semantic loss [28] to name a few). This paper, however, has a slightly different scope – we focus on leveraging ontologies (rather than general logical constraints) in neural networks. To this extent, we propose a general architecture for leveraging ontologies as

well as training techniques for it. The mentioned techniques can be used to implement one block of this architecture and we show in the paper how it can be done. Besides, we examine the human understanding of ontology-based explanations.

The rest of the paper is organized in the following way. Section II describes related work in ontology-based neuro-symbolic artificial intelligence. Section III introduces the computational framework for ontology-based training of neural networks, allowing to leverage knowledge encoded in the form of ontology to improve network accuracy, as well as use ontologies to explain the predictions of the network. Section IV describes the experimental evaluation of the proposed approach, with respect to both the prediction quality and the understandability of ontology-based explanations. Section V introduces a programming library for ontology-based explanations. Finally, Section VI summarizes the findings and discusses the perspectives of ontology-based neuro-symbolic intelligence.

II. RELATED WORK

This section briefly describes existing directions in embedding ontologies into neural networks. Typically, it is done for explainability, but some forms also contribute to the prediction quality. More specifically, these approaches can be divided into two groups: 1) ontology-aware neural architectures and training methods, and 2) post-hoc explanation techniques. Methods of the first groups contribute to both goals – they modify the prediction quality of the network and, at the same time, can be used to explain its predictions. Methods of the second group are applied to existing networks and, therefore, cannot influence prediction quality, they are designed only for building explanations.

The primary technique employed in ontology-aware neural architectures involves establishing a clear correspondence between each concept defined in the ontology and specific components (either individual neurons or groups of neurons) of the neural network. These components (typically, they are single neurons), corresponding to ontology concepts, can be internal (hidden) nodes within the network structure, as exemplified in [29], where the hierarchical nature of the neural network aligns with the ontology structure. Alternatively, and most commonly, they constitute the output layer of the neural network [18], [30], [31], [32], [33]. In the latter scenario, the network configuration resembles that typically employed for multi-label classification tasks.

The primary mechanism for establishing such correspondence between neurons and concepts lies in the formulation of the loss function. In its simplest manifestation, wherein all neurons associated with concepts are positioned in the output layer, a standard loss function, such as binary cross-entropy, suffices. Here, the alignment is achieved by requiring each sample within the training set to be labeled with all pertinent concepts, thereby directly incorporating the prediction of each concept neuron into the loss function computation.

A more intricate approach involves embedding additional logical constraints from the ontology into the loss function formulation. For instance, in [29], the loss function penalizes weights between neurons whose respective concepts are not connected in the ontology.

As a result of establishing such connections, when a sample is fed into the network, one can observe activations of these distinguished neurons (corresponding to ontology concepts) and obtain the list of concepts relevant to the sample. The result of the model itself is formed based on these concepts using some explainable method (e.g., logistic regression [30], [31], [32], [33]) or an inference engine [18]. Thus, this result becomes interpretable in terms of ontology by construction, as it is obtained through an explainable method. However, the relationship with the original input features, as a rule, is beyond the level of abstraction addressed by the respective method (refer to the introductory note on varying levels of explanation).

A less common technique involves modifying the original feature space to enhance its semantic richness, aiming to facilitate the comprehension of standard explanation methods to non-specialists. This technique is used, e.g., in [34].

Among the mentioned papers, some were focused mostly on enhancing the explainability and did not evaluate the influence of using ontologies on the network prediction quality. As for those where such evaluations were done, the results are varying. In some papers, the prediction quality of the ontology-aware architecture is slightly worse than that of the non-ontology-aware model [29], [30], [31]. Others report either the same prediction quality but achieved with fewer training samples [33] or prediction quality improvement [32], [34]. We may argue that varying results related to the prediction quality can be explained not only by the effectiveness of the techniques of leveraging ontologies but largely by the expressiveness of the ontology and its relationship with the classification target.

Post-hoc explanation techniques are, in a sense, more universal, because potentially they can be applied to any neural network, trained without explainability requirements, to enable its concept-based explainability, however, as it was noted, they do not have the potential for improving prediction quality. The primary approach to developing post-hoc ontology-based explanation methods involves creating an explainable approximation of the “black box” model. This approximating model aims to produce results similar to those of the “black box” model. Decision trees are commonly used for this purpose [35], [36], [37], [38]. For instance, in [35], [36], and [37], ontologies enhance the interpretability of decision trees. During the decision tree construction, the method evaluates potential partitions by considering the generality of concepts based on their position in the ontology, prioritizing more general concepts first. A significant issue with post-hoc explanations using approximating models is that the approximator is only indirectly related to the original model, meaning it may not accurately explain why the original model reached a particular conclusion.

A fundamentally different approach to ontology-based post-hoc explanation of neural networks, initially proposed in [39] and further developed in [40] and [41], demonstrates that the internal representations of a neural network trained on a specific problem can, under certain conditions, align with the concepts of the problem domain ontology. This alignment allows for the identification of ontology concepts associated with a sample processed by the network, enabling the formation of explanations through ontological inference based on these identified concepts. A key challenge in this approach is finding the set of neurons whose activations are most informative for extracting a particular concept. Algorithms addressing this challenge were introduced in [39] and [41].

Another important aspect in understanding the role of ontologies in neural network explainability is how the ontologies are used to construct the explanation (and what exactly is meant by an explanation).

A straightforward method for using ontologies to build explanations involves logical inference engines. Initially, a set of concepts associated with the sample is identified using one of the methods listed above. The sample is then treated as an anonymous individual within the ontology, and a set of statements is generated linking it to the identified concepts. Subsequently, logical inference is applied, which uses the sample's description and the ontology's concept definitions to determine if the concept belongs to the target class (the class for which the model is trained). The axioms and inference rules used in this process constitute the explanation [39]. For smaller ontologies and limited sets of axioms, this approach can be significantly accelerated by preparing special tensors and using a GPU [18] for inference. It is important to note that the selection of related concepts is often probabilistic, which can also be accounted for by the inference engine. Additionally, explanations can be generalized using inductive logic programming [42]. In [43], a variation of this method is employed where the target class obtained through inference is compared with the class predicted by the original model. If there is a discrepancy, the user is notified of the classification system's low confidence.

In some methods, ontologies are used indirectly to derive a set of interpretable features, enhancing the comprehensibility of any explanation method that demonstrates the influence of these features on the model's final outcome. For example, this approach can be applied in many self-explaining models where a layer of concepts is introduced, followed by logistic regression [31], [33], and other models [29], [44]. Since many self-explaining models establish a clear relationship between a neuron and a concept, methods that show a neuron's contribution to prediction (such as layer-wise relevance propagation) become more meaningful [29]. A similar principle is employed in [34], where easily interpretable semantic features (attributes) of classified objects are created.

In methods described in [35], [36], and [37], ontologies are used even more indirectly. Explanations are derived from a

decision tree, where each path from root to leaf represents an explanation. The classification of a sample using a decision tree can also be seen as an inference process, with the path from root to leaf serving as a certificate of this inference. The nodes of the decision tree are ordered according to the generality of terms defined by the ontology, which can enhance interpretability.

The authors of [45] categorize techniques for building explanations into two types: logical and attributive. Logical explanations provide a chain of logical inferences, whereas attributive explanations link the model's decision to the relevance of the sample to specific concepts. Table 1 (adapted from [45]) summarizes the existing techniques for integrating ontologies into neural networks highlighting problem setting, the technique used, and how the ontology is utilized in the explanation.

TABLE 1. Main approaches for integrating ontologies into neural networks.

Approach	Technique	Explanation principle	
		Logical	Attributive
Ontology-aware neural architecture	Concepts correspond to inner nodes of the network		[29]
	Concepts correspond to the output nodes of the network	[18], [43]	[30]–[33]
	Semantic input features		[34]
Post-hoc alignment approaches	Neural network approximation by a decision tree	[35]–[37]	
	Mapping networks (mapping activation values of the neural network to concepts)	[39], [41]	

III. COMPUTATIONAL FRAMEWORK FOR ONTOLOGY-BASED NEURAL NETWORKS

This section describes a general framework for building explainable neural networks, accounting for a domain ontology. The framework description consists of two main parts: neural architecture and the ways of ontology representation, compatible with this architecture.

The main problem the proposed framework is intended to solve is providing the set of ontology concepts relevant to the sample passed to the neural network. Here, by “relevance” we mean that the input sample (in some, problem-specific way) represents an individual of the respective concept. For example, if the ontology defines the scientific classification of plants, then relevant concepts to the image of a daisy can be *Bellis perennis* (as it is its species name), *Asteraceae* (family name), *Plant* (kingdom name), etc. Knowing the set of relevant concepts, one can produce either logical or attributive explanations, using methods listed in Sec. II.

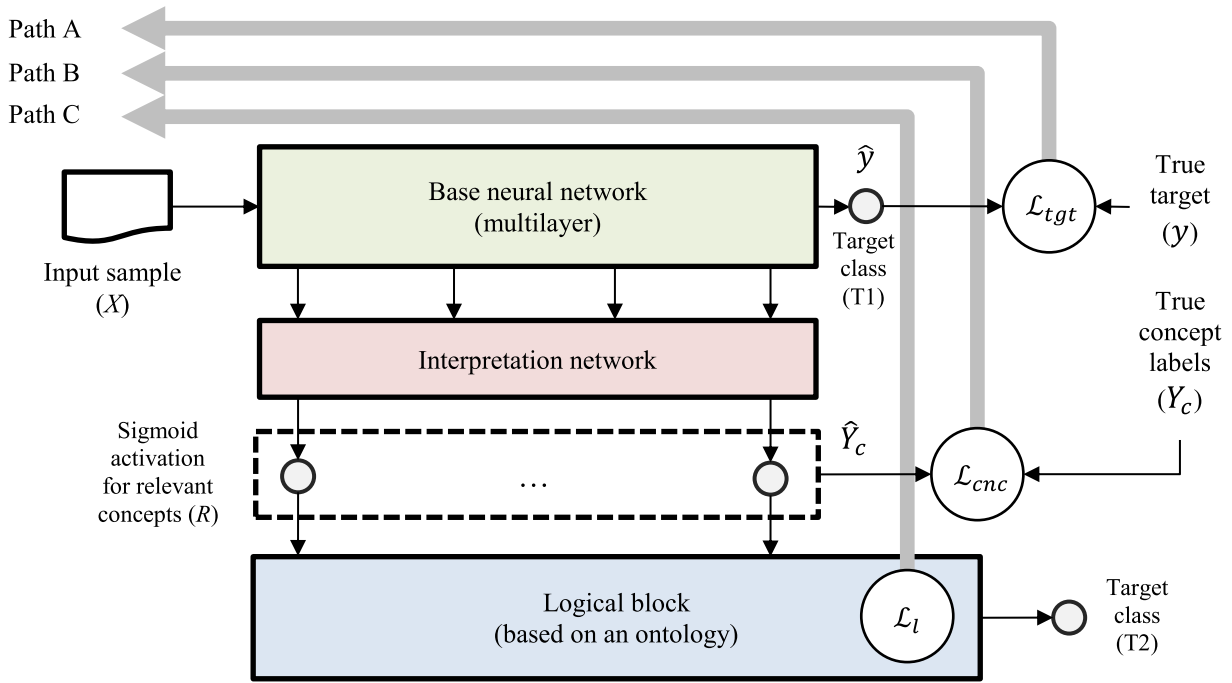


FIGURE 1. General framework for constructing explainable ontology-based neural networks.

A. NEURAL ARCHITECTURE

The core of the proposed approach is a generalized framework for constructing explainable neural networks using the ontology of the problem domain. This framework combines features of post-hoc approaches to explanation and self-explaining architectures. Based on this framework, specific self-explaining architectures of neural networks and methods for their training can be created, taking into account the characteristics of the problem, the amount of available data, and the structure of the problem domain ontology.

It should be noted that the framework is based on the assumption that for the particular problem (classification) under consideration it is fundamentally possible to form a solution based on an ontological inference (that is, there is an ontology that defines such a set of concepts and relations between them that the target classification problem can be expressed using them). This assumption is quite common for methods of logical explanation since without its fulfillment the logical explanation itself loses its meaning. The need to train a neural network (and not solve the problem exclusively using symbolic methods) may be due to the fact that the explanation does not cover all the logic connecting the input (observable) features that describe the problem, but only part of them. That is, in general, the solution to the specific problem can be carried out in two stages:

- 1) transition from observed features to symbols (ontology concepts), and
- 2) obtaining the final solution through manipulation of symbols (within the framework of one or another symbolic, logical model).

Moreover, the explanation under consideration covers only the second stage.

According to the proposed framework, the explainable neural network includes three blocks (Fig. 1):

- 1) Base neural network (consisting of several layers, i.e. belonging to the class of deep neural networks).
- 2) Interpretation network.
- 3) Logical block.

The base network takes an input sample and solves the problem by classifying this sample (for example, assigning it or not assigning it to a given class, if it is a binary classification problem). The proposed framework does not impose special restrictions on the underlying base network (except that it must be multi-layered) - it can be organized following any of the architectures that have proven their effectiveness in solving the target problem (for example, if the task is image classification, then the base network can be ResNet [46], EfficientNet [47], MobileNet [48] or some other common convolutional network).

The interpretation network receives activations of some intermediate layers of the base network, and its outputs correspond to concepts of the problem domain ontology relevant to the sample processed by the base network. The goal of the interpretation network is to establish connections between the internal representations of the processed sample built by the base neural network and human-understandable ontology concepts. The interpretation network is trainable (see below), and its training requires samples, labeled for all ontology concepts that should be placed during explanation.

The logical block takes into account the relationships between ontology concepts (for example, the incompatibility of two concepts, subclass relation, and others). These logical relationships are not learnable but are represented in a differentiable form so that they can be used in the training process of both base and interpretation networks. One of the outputs of the logical block is the target class label (T2 in Fig. 1). Note, that in this paper we do not propose an original architecture of the logical block, instead, we show how ontology statements (in description logic) can be mapped to propositional logic statements, and the latter can be encoded using some of the existing techniques (e.g., semantic loss [28]).

Formally, training of this architecture can be defined in the following way. Let O be an ontology and $\mathcal{V} = \{(X, y, Y_c)\}_n$ – a training dataset, where $X \in \mathcal{X}$ is an input sample (e.g., image, data row, etc.), $y \in \{0, 1, \Delta\}$ is the target label (we assume that the target label corresponds to some class in the ontology O), and $Y_c \in \{0, 1, \Delta\}^{|\mathcal{R}|}$ are concept labels. Here, Δ is an empty value (i.e., for some samples in $\mathcal{V}$ target value may be unknown), and $\mathcal{R}$ – the set of ontology concepts used to define target class, also called as “relevant concepts” (these, and only these concepts are used to form explanations). Base deep neural network G performs binary classification, therefore, its output typically has sigmoid activation function and $\hat{y} = G(X) \in (0; 1)$ (this framework can also be extended to the multiclass classification). We assume, that G is a multilayer network, and to establish the connection between G and the interpretation network I one has to select some subset S of layers of network G . We denote the output of some particular layer s as $G_s(X)$. Interpretation network is defined to take these values and output probabilities for each relevant concept (therefore, sigmoid activation): $\hat{Y}_c = I((G_s(X))_{s \in S}) \in (0; 1)^{|\mathcal{R}|}$.

The proposed framework allows for the propagation of error gradients and, accordingly, training of the neural network component along the following paths (indicated by arrows in Fig. 1):

- Path A – if the target class is known, the weights of the base network can be adjusted using backpropagation. Error propagation along path A represents normal training of the base network without taking into account the symbolic component. This path is implemented using binary cross entropy loss function (ℓ_{BCE}) according to (1):

$$\mathcal{L}_{tgt}(\hat{y}, y) = \begin{cases} \ell_{BCE}(\hat{y}, y), & y \neq \Delta \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

- Path B – if the set of all ontology concepts relevant to the sample is known, both the weights of the base network and the weights of the interpretation network can be adjusted using error backpropagation. The result of this is an increased correspondence between the concepts of the problem domain and the internal representations generated by the underlying network. This path is

implemented using the loss function presented in (2):

$$\mathcal{L}_{cnc}(\hat{Y}_c, Y_c) = \sum_{i \in |\mathcal{R}|} \mathbb{I}[Y_c \neq \Delta] \ell_{BCE}(\hat{Y}_{c,i}, Y_{c,i}). \quad (2)$$

- Path C – adjust both the weights of the interpretation network and the weights of the base network, based on the logical definitions. This path is based on some form of logical loss $\mathcal{L}_l(\hat{y}, \hat{Y}_c)$, typically penalizing inconsistencies in the assignment of variables to the logical formula. For example, it can be defined using the probability to sample a valid variable assignment for a given logical formula from a number of Bernoulli distributions [28] – the lower the probability, the higher the loss value. There are several ways to implement this loss, and any of them can be plugged into the proposed architecture. Note also, that this loss does not use ground truth labels, therefore, this path can be employed even for unlabeled samples.

Using (3), a specific training procedure must combine error propagation along all three listed paths (the proportion between them is determined by the specific method of forming the architecture of an explainable ontology-based network):

$$\mathcal{L} = \mathcal{L}_{tgt} + \beta_1 \mathcal{L}_{cnc} + \beta_2 \mathcal{L}_l, \quad (3)$$

where $\beta_1, \beta_2 \geq 0$ – weights of the respective components of the loss function.

It should be noted, that path B requires information about the relevance of the sample to other concepts in the problem domain (Y_c). Despite the fact that this information is rarer and its acquisition is a more labor-intensive process than labeling exclusively to the target class, it is not possible to completely eliminate path B – without appropriate training examples it turns out to be impossible to establish a connection between concepts and the outputs of the interpretation block. Accordingly, ontology-based explanations are possible only when there is at least a small number of samples labeled with all the concepts of the network (through which the target class is determined).

A useful property of the proposed framework is the ability to assess the internal consistency of predictions obtained in different ways (T1 and T2). If the base model and the logic block based on the interpretation block data produce different results, the user may be informed about the low confidence of the result. At the same time, the user can be presented with both a logically substantiated result (based, however, on not entirely reliable assessments of the relevance of the sample to the basic concepts) and the result of the basic model, but without explanation.

B. ONTOLOGY REPRESENTATION

The logical block can be constructed using one of the existing approaches allowing to take into account logical constraints and logical relations when training neural networks (NeurASP [24], DeepProbLog [25], LTN [26], STE [27],

semantic loss [28], etc.). To do this, it is necessary to transform the set of ontology axioms into a set of statements in a formal language that is supported by the appropriate approach. All of these languages rely in one way or another on first-order predicate logic or propositional logic; ontologies described using the OWL language also represent a subset of first-order predicate logic, so such a transformation does not cause any fundamental difficulties.

Let us consider, in particular, the structure of the logical block and the method of its formation, if the semantic loss function is used as a method for taking into account logical restrictions. The semantic loss function [28] allows processing only logical constraints specified in propositional logic, so the ontology (or its fragment) must be translated into the language of propositional logic (in this case, a certain loss of detail is inevitable).

1) Propositional symbols definition:

- Ontology class, corresponding to the target class of the base network becomes a propositional symbol.
- Atomic-named ontology classes, used in the definition of the target class also become propositional symbols.
- Complex classes, built using description logic constructs $\forall R.C$ (class, which individuals are connected via role R only with individuals of C , axioms `ObjectAllValuesFrom`, `DataAllValuesFrom` of OWL 2 language [49]) and $\exists R.C$ (class, which individuals are connected via role R with at least one individual of C – axioms `ObjectSomeValuesFrom`, `DataSomeValuesFrom` of OWL 2 language), also become propositional symbols, because these constructs cannot be transformed into more fine-grained propositional structures.

2) Logical connections translation:

- Class intersection ($A \sqcap B$) is translated into conjunction of the propositional symbols, corresponding to these classes ($P_A \wedge P_B$).
- Class union ($A \sqcup B$) is translated into disjunction of the propositional symbols, corresponding to these classes ($P_A \vee P_B$).

3) Axiom translation:

- Class equivalence ($A \equiv B$) is translated into propositional symbols equivalence ($P_A \Leftrightarrow P_B$).
- Class inclusion (subclass relation) ($A \sqsubseteq B$) is translated into implication ($P_A \rightarrow P_B$).
- Class disjointness ($A \sqcap B \equiv \perp$) is translated into the following disjunction ($\bar{P}_A \vee \bar{P}_B$).

After such a translation, all propositional symbols introduced in this way and relevant to the target class become separate inputs of the logical block. Based on the definitions obtained in step 3 the logical formula is built and encoded into the loss $\mathcal{L}_l$, according to the selected approach for embedding logics into the neural net.

IV. EXPERIMENTAL EVALUATION

Experimental evaluation is aimed at answering two main questions. First, how explainability affects the prediction

quality of the neural network built using the proposed framework. Second, how ontological explanations affect the user experience.

A. DATASETS

We perform evaluation on two public datasets popular in the field of ontology-based neural explanations: XTRAINS [50] and SCDB [51]. Both datasets consist of a collection of images, an ontology, and image annotations by relevant concepts of the ontology.

Explainable Abstract Trains Dataset (XTRAINS) was proposed in [50] to facilitate research on explanation and justification (it was used for this purpose, e.g., in [39], [40], and [41]). This dataset contains 500 000 synthetic images of 152×152 .

Each image contains a schematic representation (drawing) of one train on a random background (see examples in Fig. 2). A train consists of a locomotive and one to five wagons.

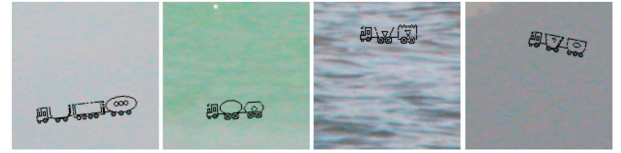


FIGURE 2. Sample images from the XTRAINS dataset.

Wagons differ by their visual appearance (e.g., number of wheels, shape, length, presence of some “load”, etc.) and based on these features they can be categorized into some classes defined by this dataset (e.g., `EmptyWagon` or `LongWagon`).

Wagon classes as well as train classes (determined by the classes of wagons making up a particular train) are defined in an ontology, accompanying the dataset (Fig. 3 shows a subset of this ontology). The ontology is described using the OWL language, which allows the representation of the subject domain using the constructs of description logic. This ontology contains 27 classes and 159 axioms (such as `SubClassOf`, `EquivalentTo`, etc.).

Train classes (and therefore, images, as one image always contains one train) are connected with wagon classes via the “has” role. E.g., `EmptyTrain` is defined as a train that can have only empty wagons (instances of `EmptyWagon`) and a locomotive and must have at least one empty wagon.

The ontology also defines a set of “artificial” concepts: `TypeA`, `TypeB`, and `TypeC`. These concepts purport to serve as classification targets of the network being explained. They are defined based on other concepts (train types), and the dataset can be used to check if an explanation technique can detect this connection between concepts and use it for explanations. This models a situation when a neural network is trained to make a classification on some complex target that, however, can be logically expressed via other domain concepts.

$$\begin{aligned}
\text{TypeA} &\equiv \text{WarTrain} \sqcup \text{EmptyTrain} \\
\text{WarTrain} &\sqsupseteq \exists \text{has.ReinforcedCar} \sqcap \exists \text{has.PassengerCar} \\
\text{EmptyTrain} &\equiv \forall \text{has.} (\text{EmptyWagon} \sqcup \text{Locomotive}) \sqcap \exists \text{has.EmptyWagon} \\
\text{Wagon} &\equiv \text{EmptyWagon} \sqcup \text{PassengerCar} \sqcup \text{FreightWagon} \\
\text{EmptyWagon} \sqcap \text{PassengerCar} &\sqsubseteq \perp \\
\text{EmptyWagon} \sqcap \text{FreightWagon} &\sqsubseteq \perp \\
\text{PassengerCar} \sqcap \text{FreightWagon} &\sqsubseteq \perp
\end{aligned}$$

FIGURE 3. Fragment of XTRAINS dataset ontology.

Each image of the dataset is annotated with a set of binary attributes, corresponding to ontology concepts. Therefore, for each image not only target label is known (e.g., whether it belongs to TypeA or not), but also the relation to all ontology concepts, which allows one to train and verify concept extraction models.

Simple Concept DataBase (SCDB) was proposed in [51] to mimic the relationships between concepts and classes arising in medical image analysis tasks, as well as to facilitate research in the field of explainable AI on localization of concepts. This dataset contains 13 500 synthetic images of size 2048×2048 .

Each of the images (see Fig. 4) contains a black background on which there is one large geometric shape, the location, rotation angle, color, and size of which are random. This shape contains smaller geometric shapes and is surrounded by them.

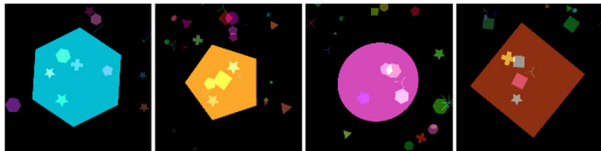


FIGURE 4. Sample images from the SCDB dataset.

Depending on the combination of shapes inside the large geometric shape, the image can be assigned to one of two classes (C1 or C2) defined in the dataset, e.g. the image is classified as C1 if there are a hexagon and a star shape inside the large geometric shape. Although no ontology was provided in the original dataset, one was created based on the known logical dependencies defining the target classes (C1 and C2). Also, inspired by the XTRAINS dataset, we introduced a “has” relation (or, role in terms of description logic), connecting the dataset image with shapes, residing in the large geometric shape. For example, $\exists \text{has.Star}$ corresponds to a class of images, where there is a star shape inside the large geometric shape. The resulting ontology consists of 18 classes and 48 axioms. The subset of the ontology defining the class C1 is shown in Fig. 5.

B. ACCURACY EVALUATION

Based on the proposed general framework, several specific methods and architectures of self-explaining neural networks

$$\begin{aligned}
\text{C1} &\equiv \text{HexStar} \sqcup \text{EllStar} \sqcup \text{TEStarmarker} \\
\text{HexStar} &\sqsupseteq \exists \text{has.Hexagon} \sqcap \exists \text{has.Star} \\
\text{EllStar} &\sqsupseteq \exists \text{has.Ellipse} \sqcap \exists \text{has.Star} \\
\text{TEStarmarker} &\sqsupseteq \exists \text{has.Triangle} \sqcap \exists \text{has.Ellipse} \sqcap \\
&\quad \sqcap \exists \text{has.Starmarker}
\end{aligned}$$

FIGURE 5. Fragment of SCDB dataset ontology.

were developed, differing both in the design of the logical block and in the method of training (using various combinations of paths A, B, and C during error propagation). It is worth noting that although the interpretation network is present in the proposed framework, in the experiments we consider a special case of such a network, namely the multi-label output layer of the base neural network. A detailed study of the capabilities of the interpretation network and its architectures was carried out in [52]. Experimental evaluation includes three following scenarios:

- 1) Training self-explaining model without logical block. In this scenario, the self-explaining model is trained with additional training data (including relevant concepts), but without a logical block.
- 2) Integration of ontological knowledge into the architecture of the self-explaining model. In this scenario, various methods of integrating knowledge into self-explaining models are considered to improve the quality of classification.
- 3) Using a semantic loss function to improve the quality of training the self-explaining model via semi-supervised learning.

In all scenarios, the ResNet-18 [46] architecture is considered. Its classification layer is replaced by a single fully connected layer containing 512 input neurons, and the number of output neurons corresponds to the number of ontology concepts, including the target class. This layer is followed by the sigmoid activation function, which forms the probability of the sample's relation to each of the ontology concepts.

No pre-trained weights were used to train the networks, that is, the weights were initialized randomly. The whole network was trained (both backbone and classification layer). The Adam optimizer with a learning rate of 0.001 was used for training. Binary cross-entropy was used as the standard loss function. The training was performed on NVIDIA GeForce RTX 3090.

Five instances of each network were trained for each case within each of the scenarios. Classification quality was evaluated using the ROC AUC metric. Classification quality distribution diagrams were plotted based on the results obtained.

From the datasets, the training set, validation set, and test set were selected. The validation set was used to estimate the value of loss during training. Sets of the following sizes were used for validation: 10 000 samples for the XTRAINS dataset, and 3000 samples for the SCDB dataset. An early stopping with a patience of 50 was used. The number of training epochs was limited to 500. A batch size of 250 was used. The training set was randomly shuffled before training started. The quality of the trained models was verified on the test set. Although cross-validation was not used the test set is fairly representative – 100 000 samples for the XTRAINS dataset, 3500 samples for the SCDB dataset.

For each dataset, its standard deviation and mean for image normalization were calculated. For the XTRAINS dataset: mean = [0.5686, 0.5106, 0.4589], standard deviation = [0.2327, 0.2189, 0.2499]; for the SCDB dataset: mean = [0.1249, 0.1251, 0.1253], standard deviation = [0.2521, 0.2527, 0.2531].

The experiments consider a subset of the ontology needed to identify the target concept. Thus, 6 concepts are considered for the XTRAINS dataset (see Fig. 3): TypeA (target class), WarTrain, EmptyTrain, ReinforcedCar, PassengerCar, EmptyWagon. For the SCDB dataset, there are 9 such concepts: C1 (target class), HexStar, EllStar, TEStarmarker, Hexagon, Star, Ellipse, Triangle, and Starmarker.

1) SELF-EXPLAINING MODEL WITHOUT LOGICAL BLOCK

This scenario explores the prospects of training the self-explaining network described earlier. We compare the quality of target class prediction by such a network with a binary classification network designed to predict the target class only. The self-explaining network, as noted earlier, predicts, in addition to the target concept, the entire subset of ontology concepts. However, it would be interesting to find out whether it is necessary to have labeling for all concepts for the whole training set or whether it is sufficient to label only some part of the training set for all concepts in order to ensure, on the one hand, that the classification quality of the target class is comparable to the binary classification network and, on the other hand, to ensure a satisfactory prediction quality for the remaining concepts.

For the XTRAINS dataset, the training set includes 20 000 images labeled with the target concept. Using this set, a binary classification convolutional network was trained to predict only the target concept TypeA. To train self-explaining networks, the same set was used, however, unlike the previous case, a certain subset was also labeled with relevant concepts (hereinafter referred to as the relevant subset). The size of the relevant subset varies from 500 to 10 000. Thus, it is also possible to estimate what size a given subset should be to achieve a classification quality (ROC AUC)

comparable to a binary classification network. Following the proposed learning procedure, the loss function in (3) has the following coefficients: $\beta_1 = 1$, and $\beta_2 = 0$ (since there is no logical block).

The results of these experiments on the XTRAINS dataset are presented in Fig. 6. The left diagram (Fig. 6(a)) shows the distribution of target class prediction quality by the binary classification network and the self-explaining network for which the size of the relevant subset was varied. With a subset size of 2000, the quality of classification of the target concept by the self-explaining network turns out to be comparable to the quality obtained by the binary classification network. At the same time, the spread of values in the first case is much lower. Further increasing the size of the subset labeled with relevant concepts improves the quality of predictions, and at a size of 5000, the self-explaining network outperforms the binary classification network in predicting the target class. The right diagram (Fig. 6(b)) shows the distribution of the prediction quality of the entire set of concepts (including the target class) by the self-explaining network. It can be observed that a high prediction quality (> 0.95) of the entire set of concepts is achieved when the size of the relevant subset is greater than 2000.

Similar experiments were conducted using the SCDB dataset. In this case, the training set consisted of 8000 images labeled with the target concept C1. The size of the relevant subset ranges from 500 to 8000. The experimental results using the SCDB dataset are presented in Fig. 7. In this case, the quality of classification of the target concept by the self-explaining network turns out to be comparable to the quality obtained by the binary classification network, with a subset size also equal to 2000. With the size of the relevant subset equal to 4000, the self-explaining network is superior in quality to the binary classification network.

To summarize, in order to ensure high classification accuracy of a self-explaining network, it is not necessary to use a training set fully labeled with all concepts (it is sufficient to label some part of it), but it is important that it be labeled with a target concept that is logically inferable from the relevant concepts.

2) SELF-EXPLAINING MODEL WITH LOGICAL BLOCK

To realize the logic block, methods belonging to different groups in the classification from [53] were used. So, the following approaches to organizing the logical block were considered:

- The KENN (Knowledge Enhanced Neural Networks) method [54] allows one to inject prior logical knowledge into a neural network by adding a layer on top of it that modifies the initial predictions according to the knowledge. The peculiarity of this neuro-symbolic structure is that knowledge is used not only during model training but also during inference. In this research, KENN acts as a representative of a relatively large class of methods,

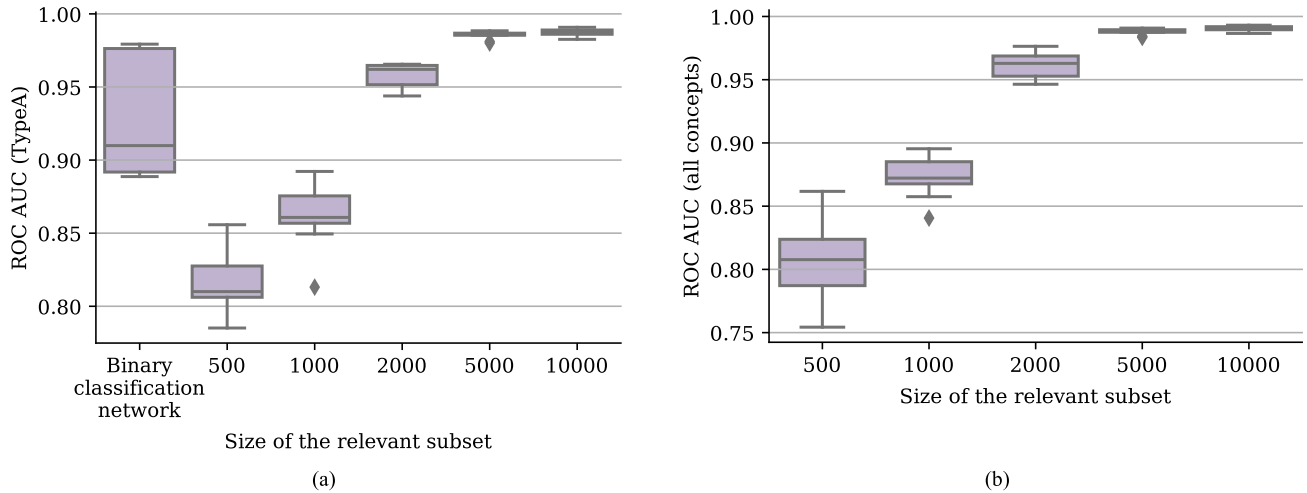


FIGURE 6. (a) Quality of prediction of the target class by the binary classification network and the self-explaining network at different sizes of the relevant subset for its training. (b) Quality of predicting the entire set of concepts by the self-explaining network. Results for the XTRAINS dataset.

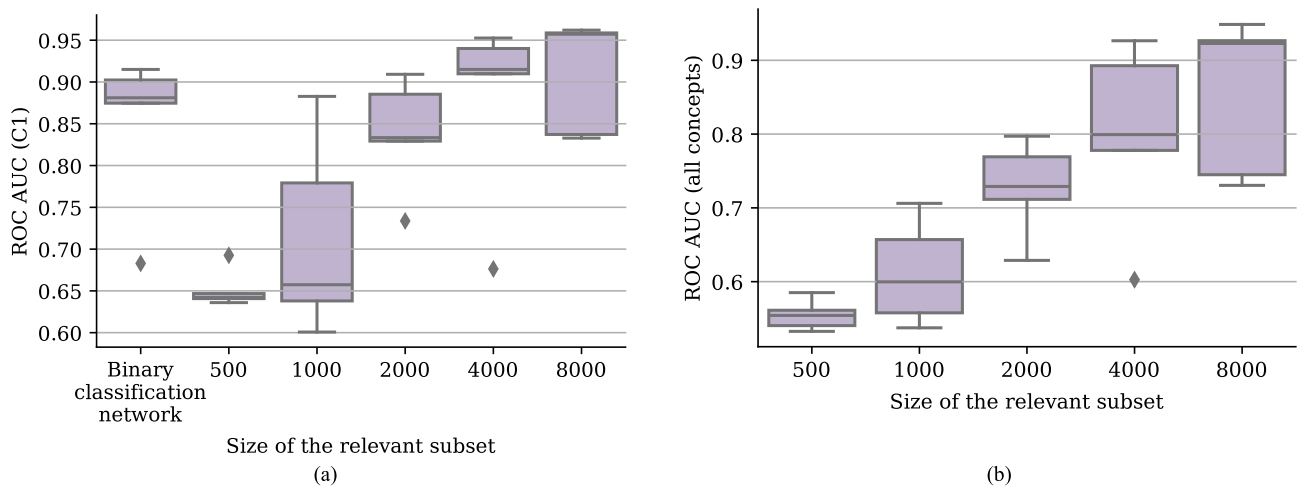


FIGURE 7. (a) Quality of prediction of the target class by the binary classification network and the self-explaining network at different sizes of the relevant subset for its training. (b) Quality of predicting the entire set of concepts by the self-explaining network. Results for the SCDB dataset.

based on fuzzy logic semantics (another representative of this class would be logical tensor networks [55]).

- The C-HMCNN (Coherent Hierarchical Multi-Label Classification Networks) approach [56] imposes hierarchy constraints on the output classes of a multi-label network. C-HMCNN is based on two main elements: a constraint layer built on top of a neural network, and a loss function that trains the model to understand when to use a prediction about the lower classes of the hierarchy to make predictions about the upper ones.
- NeurASP [24] is a neuro-symbolic approach that integrates neural networks with Answer Set Programming (ASP) to combine the strengths of learning from data and reasoning with logic. It consists of two main components: a neural network providing probabilities for atom predictions, and an ASP program imposing symbolic constraints. The model leverages neural predictions to

guide logical inference and uses ASP constraints to ensure the neural predictions satisfy logical rules during training. This setup allows for better interpretability and handling of complex reasoning tasks.

Using each of the listed methods, 5 instances of the self-explaining network were trained (the spread of results turned out to be quite small, so 5 copies are enough to make reasonable conclusions). The results of assessing the quality of networks (ROC AUC) are presented in Fig. 8. It can be seen that minor improvements are observed only when using the KENN approach. The use of C-HMCNN leads to quality degradation, which can probably be justified by the fact that the ontology fragments chosen for the study are characterized by low expressiveness. The use of NeurASP also leads to a relatively low classification quality, and at the same time is characterized by a large spread of values.

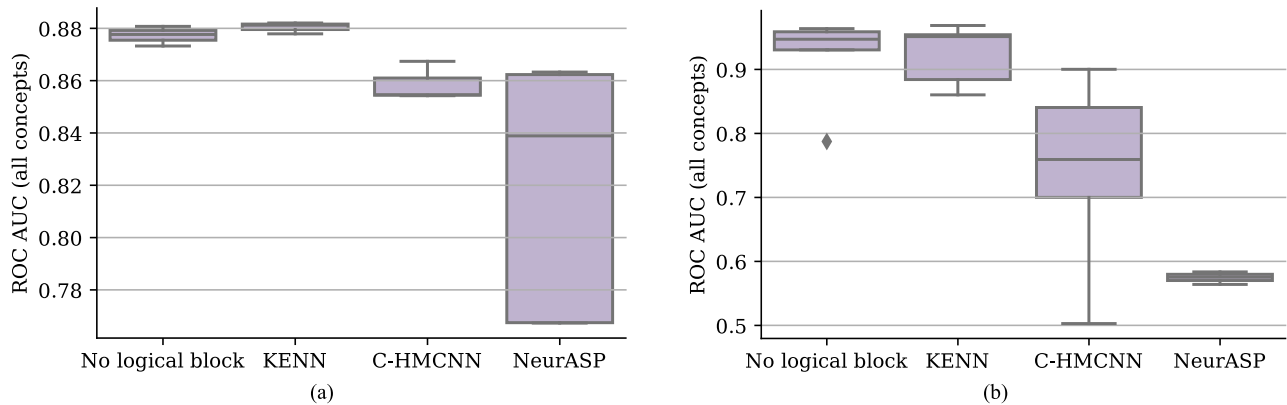


FIGURE 8. Classification quality of self-explaining models with logical block. (a) XTRAINS dataset. (b) SCDB dataset.

3) SEMI-SUPERVISED LEARNING USING SEMANTIC LOSS FUNCTION

This series of experiments investigated the impact of using a semantic loss function [28] on the quality of self-explaining models following the architecture characteristic of multi-label classification. Since the relationships between the outputs of such self-explaining networks are explicitly specified in the problem domain ontology, the semantic loss function can be considered as an additional regularizer that strengthens the relationships between concepts (network outputs).

The semantic loss function is most effective in semi-supervised learning tasks, where, in addition to the labeled data, unlabeled data are also used for training, usually in much larger quantities than the labeled data. In the experiments conducted, the unlabeled data were used to calculate the value of the semantic loss function after each training iteration (epoch), while the labeled samples were used to calculate the value of the binary cross-entropy. Thus, the proposed loss function has the following coefficients: $\beta_1 = 1$, and $\beta_2 > 0$ (since the logical block in this case is realized using the semantic loss function). In the context of this series of experiments, we will refer to β_2 as the semantic loss weight.

The semantic loss weight is an important factor that determines the training quality and ultimately the classification accuracy of a neural network model.

The experiments were conducted using the XTRAINS dataset. In all experiments, the training set included 5000 labeled images. The number of unlabeled samples ranged from 0 to 20 000.

In the first experiment, the semantic loss weight changed as the number of training epochs passed increased. Fig. 9(a) shows the results of an experiment in which the initial semantic loss weight was assumed to be 0.1. For each subsequent epoch, the semantic loss weight was increased by a value inverse to the total number of epochs (300). In this case, the use of the semantic loss function has a positive impact on the training of the networks. For example, as the number of

unlabeled samples increases, the spread of network quality decreases. In addition, the quality of the networks is more often higher. When 20 000 unlabeled samples are used, the network slightly outperforms the baseline model in terms of classification accuracy.

Fig. 9(b) shows a diagram of the quality distribution of the networks obtained during the experiment, in which the initial semantic loss weight was 0.5. In contrast to the previous case, each subsequent epoch the semantic loss weight was reduced by a value inverse to the total number of epochs. At each iteration, the final semantic loss weight was defined as the maximum between zero and the current semantic loss weight. It can be seen that the results obtained are very similar to those obtained in the first experiment. In this case, the use of the semantic loss function has no positive effect on the training of self-explaining networks.

Summarizing the experiments conducted using the semantic loss function, we can conclude that its use can improve the quality of training and classification of self-explaining models. However, as the experiments have shown, it is more appropriate to use a small weight of the semantic loss in the first iterations of training. Otherwise, the network becomes more prone to overfitting, and the final classification accuracy may be worse than when using supervised learning.

C. EXPLANATIONS UNDERSTANDABILITY

We also conducted an experiment to investigate users' interactions with ontology-based explanations of neural networks to address two key questions:

- Do ontology-based explanations of a neural network simplify the task of the decision-maker?
- Do different forms of ontology-based explanations vary in understandability and ease of use? If so, what features contribute to this variation, and which form is preferred by users?

The first question aims to assess whether leveraging domain concepts simplifies decision-making by providing easy-to-understand cues. The second question seeks to guide the optimization of explanations. If certain ontology

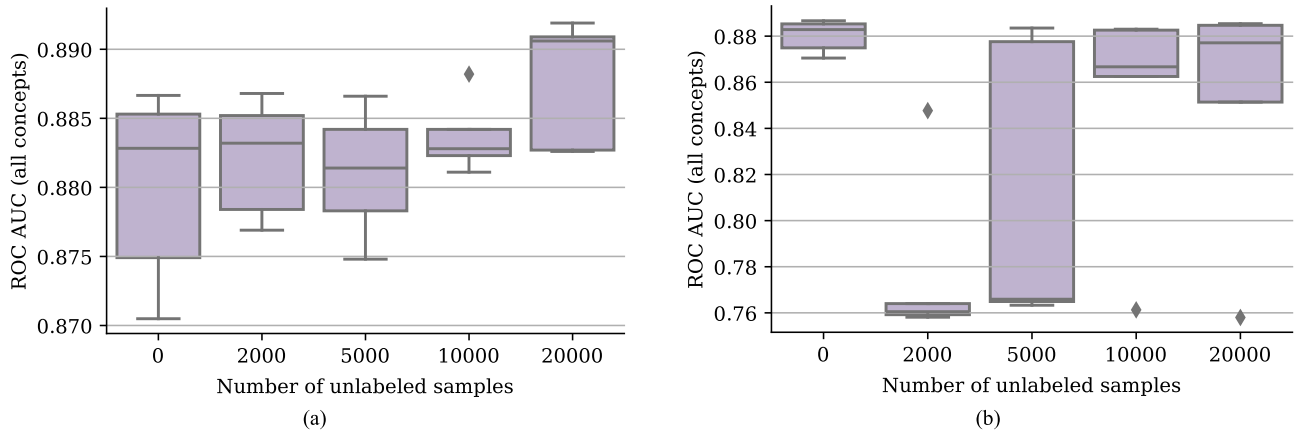


FIGURE 9. Influence of the number of unlabeled samples on the classification quality of self-explaining networks in semi-supervised learning. (a) The initial semantic loss weight is 0.1. (b) The initial semantic loss weight is 0.5.

properties significantly affect the understandability of explanations, they can be adjusted to enhance clarity and usability.

In the experimental study, we collected both objective performance metrics (such as decision time and accuracy) and subjective feedback through questionnaires.

We examined two main approaches to representing ontology-based explanations (see Sec. II):

- **Attributive explanation** (see Fig. 10): These explanations highlight the relevance of certain ontology concepts to the model output. This relevance can arise from semantically-loaded input features corresponding to concepts [34], or from a semantically-loaded pre-output layer of the network, followed by a linear classification model (e.g., logistic regression – in such a model coefficients corresponding to the concepts can be used to understand the effect of these concepts on the classification result) [31]. For example, the explanation for a sample from the XTRAINS dataset in Fig. 10 visualizes probability diagrams of its belonging to each of the ontology concepts relevant to the corresponding target class. The probability of the concept WarTrain (relevant to TypeA) is close to 1.0. At the same time, the probabilities of the concepts ReinforcedCar and PassengerCar (relevant to WarTrain) are also close to 1.0. Consequently, we can say that, according to the explanation, WarTrain is represented in the sample and, therefore, the sample belongs to TypeA.
- **Logical explanation** (see Fig. 11): These explanations demonstrate a logical inference process, showing how the target class can be deduced from a set of concepts. Logical explanations are employed, for instance, in [39]. The first sentence in each of the explanations contains the concepts found in the sample. The rest of the sentences can be considered as a chain of reasoning (ontology axioms) based on the found concepts and leading to the target class.

In this paper, we compare the understandability of explanations constructed using these two approaches.

Two main approaches to formalizing and evaluating model interpretability in user studies are commonly employed. The first approach is associated with simulatability, which refers to the user's capability to execute a model and obtain the correct result, independent of the computing device [57]. The other approach, known as “what if” explainability, pertains to a user's ability to determine the effect of small perturbations in the model's input data on its outcome [19]. In this study, we adopt an approach akin to simulatability, wherein users are tasked with following a logical definition using various concept-based cues.

To conduct the experiment, we developed a web application designed to present tasks and explanations to participants and collect their responses. The application also offers assistance to simplify tasks for participants. Specifically, the welcome screen provides detailed definitions of each ontology concept along with examples of corresponding images. Additionally, brief concept definitions can be accessed on demand.

The experiment involved 28 students from a higher educational institution aged between 20 and 24 years. Informed consent was obtained from all the participants. Each participant was tasked with completing 48 classification tasks related to trains. The number of tasks without explanations, as well as those with each type of explanation, was 16 for each case. Samples for classification were selected to include 6 instances for each combination of train labels. These samples were evenly distributed, ensuring an equal number of samples with trains (from XTRAINS dataset) of each parameter combination in each type of task (i.e., 6 samples not belonging to any type, 6 belonging only to Type A, 6 belonging only to Type B, and so forth). Tasks with different types of explanations were randomly presented to users.

The study comprised three distinct stages:

- **Introduction:** During this initial stage, participants were tasked with acquainting themselves with the subject area's description. They were required to understand the characteristics of each type of wagon and discern the

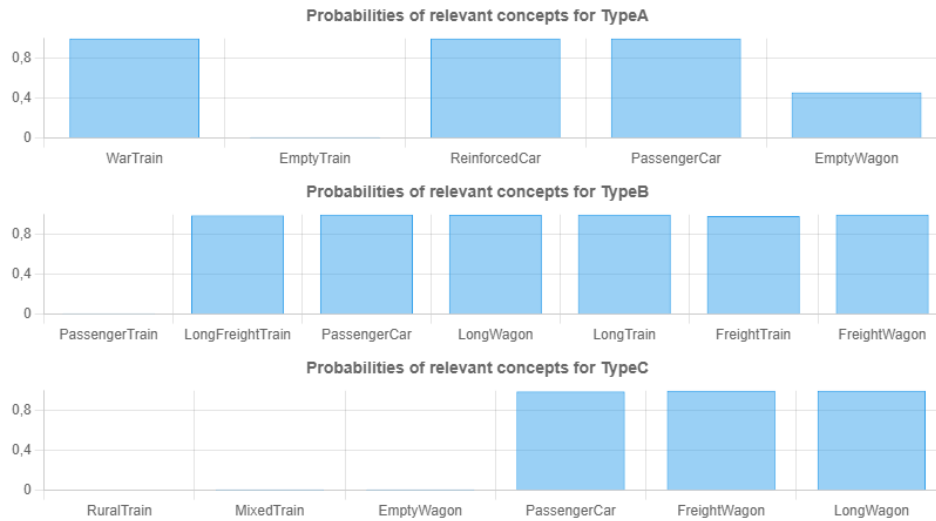


FIGURE 10. Example of an attributive explanation.

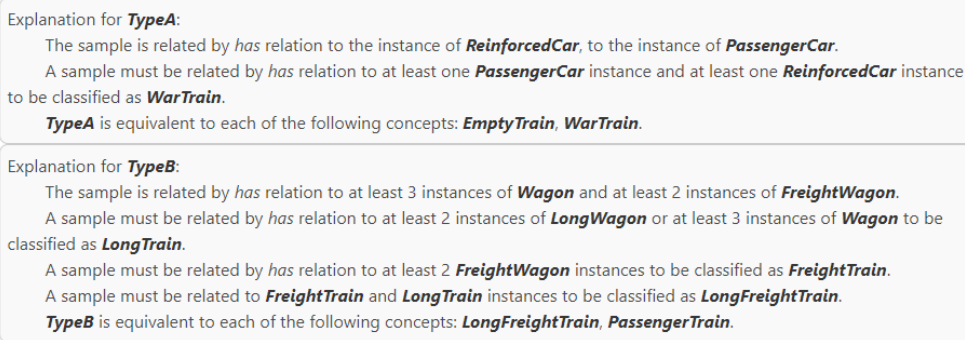


FIGURE 11. Example of a logical explanation.

combinations of wagons that define a particular type of train. Additionally, details regarding the experiment, such as the number of questions, were provided.

- **Classification:** In this stage, participants are directly engaged in classifying the presented trains. Their decision time and responses were recorded for each question.
- **Questionnaire:** Following the classification of all samples, participants were presented with a questionnaire. The purpose of this questionnaire was to solicit subjective evaluations, including which type of explanation they found easier to comprehend and which was deemed most useful for decision-making.

The questionnaire comprised questions aimed at assessing the subjective complexity of classification when utilizing each type of explanation. Subjective complexity was measured on a scale ranging from 1 to 7, where 1 denoted low complexity and 7 indicated high complexity. Additionally, participants were questioned on their utilization of the provided explanations.

During data analysis, attention was paid to participants potentially not engaging earnestly with the study.

Consequently, 8 individuals were excluded due to excessive decision times or low accuracy, suggesting a lack of understanding or effort. Hence, 20 participants were selected for further analysis, and single decision time outliers exceeding 200 seconds were eliminated.

The recorded user responses and decision times revealed that the initial 5 questions generally required more time and effort as participants familiarized themselves with various question types and explanations. Consequently, distribution parameters of objective metrics were calculated, accounting for this initial learning phase. Notably, answering questions with explanations generally took less time and yielded higher accuracy (Fig. 12).

Comparing attributive and logical explanations, classification with attributive explanations typically took less time due to their interactive and visual nature (see Fig. 12(a)). However, excluding the first 5 questions of each type, decision times for attributive explanations were comparable to those for logical explanations. Logical explanations, although more time-consuming, provided a structured reasoning sequence, reducing the likelihood of incorrect

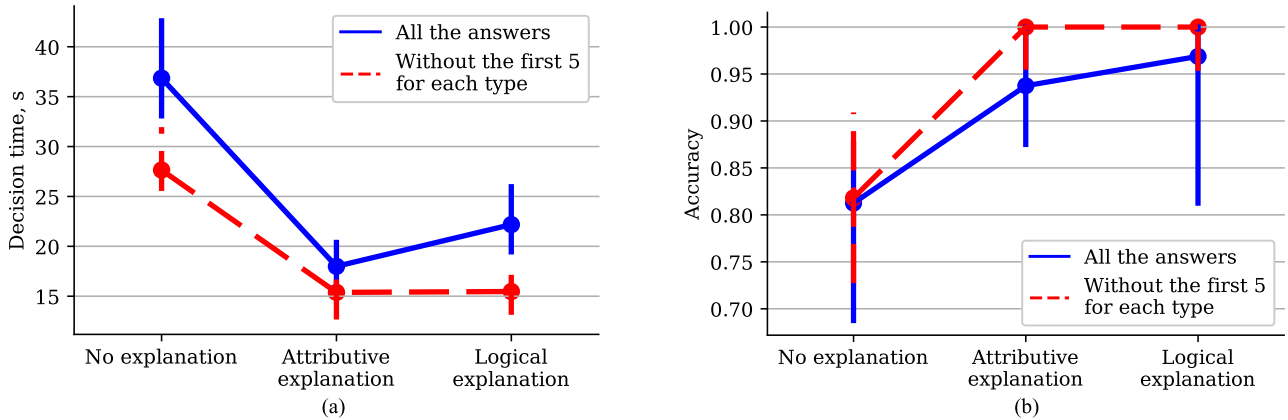


FIGURE 12. (a) and (b) Distribution of response time and accuracy, respectively, using a median estimate.

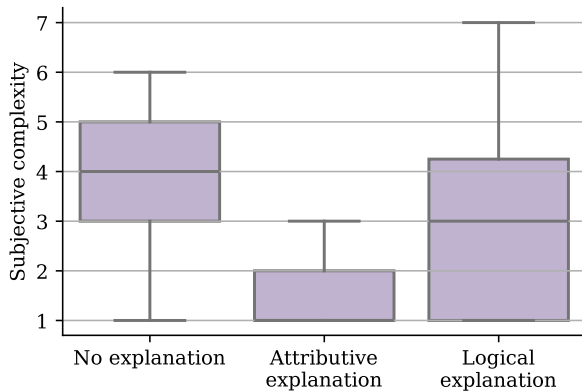


FIGURE 13. Distribution of subjective complexity of explanations.

classifications (see Fig. 12(b)). Furthermore, the median accuracy for questions with explanations, excluding the first 5 questions, was 1.0.

Regarding subjective metrics, the subjective complexity of classification was assessed on a scale of 1 to 7. Results suggested that classification without explanations posed the greatest difficulty (see Fig. 13). Attributive explanations were generally more comprehensible to users than logical ones, with the latter presenting structured text obtained from ontological inference, necessitating more time for user familiarization.

Additionally, the questionnaire collected information on various properties of explanations using the Explanation Satisfaction Scale proposed in [58]. Fig. 14(a) illustrates a radial diagram reflecting median evaluations of attributive and logical explanations for each explanatory property. Notably, attributive explanations surpassed logical ones in all aspects except completeness.

Lastly, participants' approaches to classification with explanations were investigated. The distribution of responses (Fig. 14(b)) revealed that over half of the participants utilized explanations "with caution," either conducting the classification themselves and subsequently verifying correctness or

acquainting themselves with the explanation before verification. Only 12.5% of respondents fully trusted the provided explanations.

V. PROGRAMMING LIBRARY FOR ONTOLOGY-BASED EXPLANATIONS OF NEURAL NETWORKS

In order to simplify and promote experiments with ontology-based explanations of neural networks we designed and published RevelioNN library [59]. The library implements both self-explaining methods proposed in this paper and various post-hoc ontology-based explanation methods described in the literature [41], [52]. Besides, it is easily extensible via well-documented interfaces which simplifies implementing new methods (e.g., for research purposes).

The library is built upon the PyTorch framework. In particular, it can be used to generate ontology-based explanations for existing PyTorch models (trained by a third party), as well as define new PyTorch models and use methods introduced in Sec. III to train them.

The main functions of the RevelioNN library are:

- Training of the mapping networks (as required by post-hoc methods).
- Training of the self-explaining network of the user-defined architecture.
- Based on the extracted concepts, build explanations as a logical inference (using an external reasoning module).
- Connect logical explanations with visual ones (for convolutional neural networks). This can be used to highlight image regions related to certain ontology concepts, allowing us to obtain a deeper insight into the classification logic behind the network. Visual explanations are formed using the occlusion method [23].

For building logical explanations the library uses reasoning module BUNDLE [60]. BUNDLE provides reasoning for probabilistic ontologies through the use of various reasoners. It uses the DISPONTE approach, where each axiom is assigned a probability value indicating the likelihood that a certain axiom will be accepted as true.

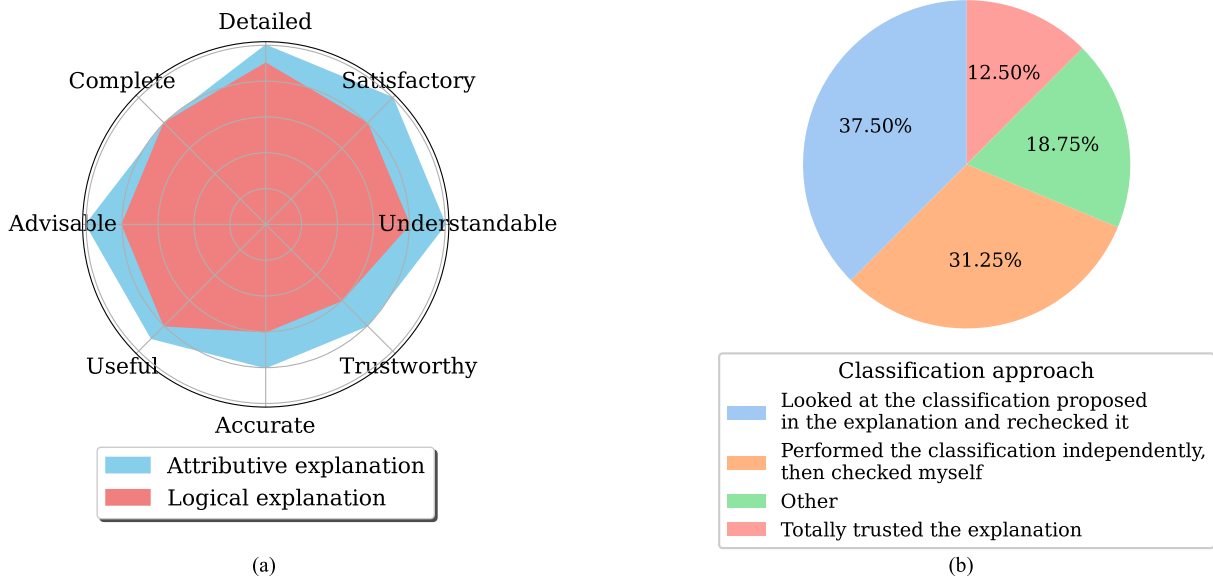


FIGURE 14. (a) Multi-criteria assessment of explanations. (b) Way of using the explanations.

The source code of the library is available on GitHub.¹ The library can also be installed from the PyPI repository of Python packages.

VI. DISCUSSION

Ontologies represent one of the promising ways of increasing the explainability of neural networks. In a broader scope, ontology-based methods of neural network explainability can be categorized as neuro-symbolic – indeed, ontology is just a way a logical system (of a special purpose). There are two main reasons why ontology-based explainability requires special attention. First is technical – ontologies (and description logics they are based on) are specifically designed with a classification task in mind (instance to class), which nicely matches the popular kind of supervised learning models, besides, ontologies are typically built “around” simple taxonomy, which is easy to reason about. The second is more pragmatic – there are many existing conceptualizations expressed as ontologies that potentially can be reused in such neuro-symbolic systems.

The proposed computational framework unifies self-explaining and post-hoc techniques for ontology-based neural networks. For example, when the weights of the base neural network are frozen and only two of the three considered gradient flow paths are used – B and C (Fig. 1), then the framework corresponds to post-hoc explanation techniques, based on concept extraction [39], [41]. When all the network weights are trainable, and there are training samples for paths A and C, then it is the self-explaining network, akin to those proposed in [18] and [43]. This general framework allows for various combined training scenarios when there are not enough fully labeled samples (labeled with all relevant concepts).

Computational results show that increasing the size of the fully-labeled dataset increases the concept extraction quality (which is expected). Besides, it also shows that combining training for the target class with training for some relevant ontology concepts (related to the target class) can improve the prediction accuracy for the target class. It is interesting because in this case, explainability turns out to be not in conflict with prediction quality. Embedding logical definitions of concepts to the neural network training does not consistently improve the prediction quality (in the experiments, it improved the quality in one dataset of the two). This might be related to the complexity of the ontology or some of its structural properties and requires further research.

The experiments on user perception of the explanations have shown that ontology-based explanations, in general, can improve the performance of the decision-maker. However, performance gain may significantly depend on the form used to convey the explanation to the user. For example, the experiment results suggest that unconditionally showing the full inference tree is not necessarily the most effective presentation form, simply highlighting ontology concepts relevant to the sample under consideration can be more effective. The likely reason is that the full inference tree contains significantly more information, which requires analysis (the role of explanation size in understandability is explored, for example, in [61]). The tradeoff between explanation completeness and its ease of use may be in interactive explanation representations, but this also requires further research.

We should also point out several limitations of the study. The proposed explanation techniques rely mostly on the ontology “core”, formed by the concept hierarchy. This limitation stems mostly from the translation process transforming the ontology (or, more precisely, its fragment relevant to the

¹<https://github.com/cais-lab/revelionn>

target class) to the explanation terms. This might be sufficient, especially if the ontology is mostly a concept hierarchy (like OWL Lite ontologies), but may also be very limiting.

Besides, the principal assumption the work is based upon is that the neural network classification target can be expressed via some combination of ontology concepts. On the one hand, it is very reasonable – for ontology-based explanations to be useful, the ontology should provide a relevant vocabulary to explain the decision. On the other hand, it significantly limits the application scope of ontology-based techniques as such. However, for the attributive explanations, this assumption is somewhat softer. Ontology concepts in this case might provide some cues, not an explanation in a strict sense.

Another possible limitation is related to the size of the ontology. In this paper, we used two rather small ontologies designed primarily for research purposes – XTRAINS and SCDB. Real-life ontologies (e.g., those, used in biology and medicine) can consist of hundreds or even thousands of concepts. The size of the ontology (or, to be precise, the number of ontology concepts, relevant to the target) influences the proposed general model in two ways. First, it influences the number of interpretation network outputs (and, therefore, the number of its parameters and its computational complexity). Second, it influences the complexity of the logical block. From the computational perspective, the first influence does not seem to be a severe issue, growth of the number of parameters depends on the interpretation network architecture, and in some architectures may be linear w.r.t. the number of concepts. However, it creates a practical burden, as for path B the samples must be labeled with all concepts, and it might be problematic if there are too many of them. The computational complexity of the logical block increases with the growth of the number of concepts much faster; some existing approaches for implementing the logical block cast the problem to a #P-complete one, therefore, are exponential w.r.t. the number of concepts. However, one does not have to use all the relevant concepts of the very large ontology. The number of concepts defines the level of explanation details, therefore, irrespective of the complexity of the whole ontology, one can define (a limited) explanation vocabulary and use the proposed approach.

VII. CONCLUSION

The paper explores the potential of using ontologies to enhance the quality and explainability of neural network classifiers. It proposes a computational framework for constructing self-explaining ontology-based neural networks. The framework unifies two branches of research in ontology-based explainability – self-explaining networks and post-hoc explanation techniques. It defines the ontology processing method, the general structure of the neural network, and several gradient propagation paths allowing one to adapt the framework to varying datasets and tasks, requiring explainability.

The proposed framework is used to construct several ontology-based self-explaining neural networks, using

existing methods of embedding logical constraints (KENN, C-HMCNN, NeurASP) and varying training models (fully-supervised training and semi-supervised training). The experimental evaluation has shown the versatility of the framework and quality improvements that can be reached by leveraging ontologies (at least, in certain problems).

The paper also explores a user perspective on ontology-based explanations. The user study has shown that using ontologies during explanations can improve the decision-maker's performance, a result that is consistent with earlier research. However, a form of presenting an ontology-based explanation can also influence the decision makers' performance – fragments of ontological inference are not always a better representation, probably because they are too detailed and may require significant cognitive effort to follow.

Future research aims to apply these techniques to a broader range of real-life ontologies, as well as to understand what ontology properties mostly influence the prediction quality and understandability gains of ontology-based neural networks.

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